Hierarchical Recovery in an Autonomous Pick-and-Place Pipeline: An Experimental Evaluation on a UR5e/2FG7 Manipulator

Arianna Bardoni
University of Genoa
Research Track II

October 3, 2026

1 Introduction

Autonomous robotic manipulation requires robust error handling to navigate real-world execution uncertainties, such as motion planning failures, object slippage during grasp, or unreachable spatial targets. Standard recovery strategies—such as local retries or resetting to a default home configuration—are frequently implemented in industrial pipelines. However, their specific statistical contributions to task robustness and their associated temporal overheads are rarely quantified under controlled, repeatable conditions.

This work evaluates a three-level recovery pipeline originally developed for the Cognitive Architectures for Robotics (CogAR) course (Assignment B1b). The architecture integrates local motion retries with planner fallback (Level 1), kinematic grasp verification with automatic retry (Level 2), and a global safe retreat maneuver (Level 3). In this report, the pipeline is evaluated across three representative experimental scenarios ($N = 5$ trials per scenario) to quantify task success rates, safe terminal state probability, and total execution time overheads.

The remainder of this report is organized as follows: Section 2 classifies the relevant literature; Section 3 outlines the research questions and hypotheses; Section 4 presents the system architecture and recovery levels; Section 5 details the experimental setup; and Sections 7–8 present the statistical methodology and experimental evaluation.

2 Related Work

2.1 Search protocol

Candidate papers were located with a general-purpose search engine (which also indexes arXiv, IEEE Xplore, ACM, Frontiers and Springer content) using queries such as *"grasp failure detection recovery"*, *"recovery behaviour manipulation"*, *"behavior tree robot recovery"*, *"motion planning fallback"*, *"robot manipulation benchmark statistical"*, *"error recovery robotic assembly"*, and *"Cartesian vs sampling-based motion planning"*. Fifteen papers were kept, and each one was checked against the publisher's page (Frontiers, arXiv, IEEE Xplore, Springer, NeurIPS, Elsevier) to confirm the real authors and venue before citing them.

2.2 Detecting and recovering from a failed grasp

In-situ failure detection and dynamic correction are critical for robust manipulation. Zhu et al. [1] combine visual perception with internal gripper telemetry to detect slip events during

pick-and-place execution, demonstrating a reduction in overall completion time relative to non-adaptive baselines. Burgess-Limerick et al. [2] introduce intentional, controlled failure injection during task execution (an artificial delay before a grasp succeeds) to measure recovery time under varying task difficulty levels, inspiring the controlled probability model (p) utilized in this benchmark. Li et al. [6] combine tactile and visual signals to detect slip during grasping, a classical result showing that a physical, sensor-based signal of grasp failure is achievable with high accuracy; our pipeline instead relies on a kinematic test-lift as a coarser proxy for the same signal. Rodriguez et al. [7] formalize the trade-off between the time cost of retrying a grasp and the reliability gained by doing so, and propose aborting a grasp attempt early once it looks unlikely to succeed, which is conceptually close to our Level 2 retry.

2.3 Organising recovery behaviours in levels

Structuring recovery actions into clear functional hierarchies simplifies error handling in multi-stage tasks. Wu et al. [3] organize a manipulation task as a graph of steps and, when an anomaly such as an external collision is detected, revert execution to an earlier step chosen from a state-dependency chart, rather than aborting the task; this matches the idea behind our local retries (Levels 1–2) versus a full reset (Level 3). Ahmad et al. [4] apply Behavior Trees (BTs) to build modular, reactive error handling. Similarly, Liu et al. [5] propose a taxonomy classifying recovery actions by structural complexity.

2.4 Recovery in assembly and industrial tasks

A related line of work looks at recovery in industrial assembly rather than pick-and-place, but faces the same core problem: deciding what to do once a failure is detected. Hayami et al. [8] predict the type of error in a snap-fit assembly from force/torque signals and then execute a recovery motion chosen for that specific error, showing that recovery can be tailored to the failure rather than being a single fixed behaviour. Chang and Kulić [9] represent a task as a Petri net and let the robot recover from known errors automatically, or learn a new recovery strategy from a human demonstration when the error is not already represented in the net. Matsuoka and Sawaragi [10] go further and plan the recovery action itself based on the meaning of the failure and on how much time is left before a deadline, which is a more flexible alternative to our fixed, always-the-same recovery sequence.

2.5 Retrying or switching planner when motion planning fails

When straight-line Cartesian path planning fails due to workspace constraints or kinematic singularities, falling back to a different, more general planner is a common way to still complete the task. This is exactly what our Level 1 does (Cartesian first, then OMPL), and it is the strategy described in MoveIt task constructor tutorials [13]. Two more academic works give this idea real experimental support: Sotirchos and Ajanović [11] directly benchmark a sampling-based planner (OMPL's RRT-Connect) against a search-based one on the same planning problems, finding that neither is always better, which supports our choice of trying the cheap planner first and only falling back when needed. Fidalgo Astorquia et al. [12] use a very similar robot to ours (a UR3e arm with an OnRobot gripper) to compare classical OMPL planners against a learned planner on success rate and planning time, the same two quantities we track for our Level 1 fallback.

2.6 How these systems should be evaluated

Rigorous comparative evaluation requires going beyond basic mean metrics. Dasari et al. [14] demonstrate that reporting average execution times without statistical hypothesis testing can yield misleading conclusions in small-sample manipulation experiments. Furthermore, Norton

and Flynn [15] provide guidelines on trial repetitions and environmental repeatability, motivating our choice of sample size and paired trial design.

2.7 What is still missing

While the existing literature addresses isolated error recovery techniques or general architectures, there is a lack of systematic quantitative studies evaluating the isolated success contribution and time cost of hierarchical recovery levels across distinct failure scenarios. This study addresses that specific gap.

3 Research Question and Hypotheses

Research Question. Does the integration of hierarchical recovery levels ($L1, L2, L3$) in a UR5e/2FG7 pick-and-place pipeline significantly increase task completion rates and safe terminal state outcomes under execution failures, and what execution time overhead is incurred by each level?

All hypotheses are evaluated at a significance threshold of $\alpha = 0.05$.

H1 (Grasp Recovery Success): Local grasp retries (Level 2) allow 100% recovery from detected grasp verification failures without triggering task aborts.
H_0: Task success rates under forced grasp failures are equal to zero ($p_{success} = 0$). H_1: Task success rates under Level 2 recovery are significantly higher than zero.

H2 (Execution Time Overhead): Resolving a failed grasp attempt via Level 2 local retries introduces a measurable temporal overhead relative to nominal execution.
H_0: Total execution time is invariant to the activation of Level 2 retries ($T_{retry} = T_{nominal}$).
H_1: Total execution time increases significantly when Level 2 retry loops are executed ($T_{retry} > T_{nominal}$).

H3 (Global Safety Rate): Upon encountering an unrecoverable planning failure (e.g., unreachable target), Level 3 (Safe Retreat) ensures that the manipulator terminates in a collision-free safe home state.
H_0: Safe terminal state rate under unrecoverable failure is equal to zero. H_1: Safe terminal state rate under Level 3 retreat is significantly higher than zero.

4 System Under Study

4.1 Pipeline

The experimental platform is simulated using ROS 2 Humble and MoveIt 2, incorporating a Universal Robots UR5e arm equipped with an OnRobot 2FG7 parallel gripper. The pipeline operates under kinematic constraints without contact dynamics modeling. The target task consists of picking a cylindrical object and relocating it to a designated drop zone. The execution sequence comprises: pre-grasp positioning, descent, jaw closure, kinematic verification via test lift, transfer, object release, retreat, and return to the home configuration. Primary movements utilize straight-line Cartesian motion planning.

4.2 The three recovery levels

Level 1 – Motion Retry: Re-attempts failed Cartesian trajectories up to 3 times. If Cartesian interpolation fails, the system falls back to a general sampling-based joint-space planner (OMPL/RRTConnect).

Reference	Topic	Failure/problem	Solution proposed	Relation to our work
[1]	Grasp failure recovery	Occluded/failed grasp	Camera + gripper cues, on-line correction	Same problem (grasp recovery), different sensors
[2]	Grasp failure recovery	Injected grasp delay	Retry while moving	Same idea of injected failures with a controlled parameter
[6]	Grasp failure detection	Slip during grasp	Tactile + visual deep-learning classifier	Physical alternative to our kinematic test-lift signal
[7]	Grasp failure recovery	Time cost of blind retries	Early abort once success looks unlikely	Basis for the retry-vs-time trade-off behind Level 2
[3]	Recovery architecture	Multi-step task failure (external disturbance)	Revert to an earlier step chosen from a state-dependency chart	Basis for our L1/L2 vs. L3 split
[4]	Recovery architecture	Adaptable recovery	Behaviour trees + learning	Alternative architecture to our fixed-order retries
[5]	Recovery classification	No shared way to name recovery levels	Levels of recovery actions	Used to justify our system recovery levels
[8]	Assembly recovery	Snap-fit assembly failure	Predict error type, then apply a matched recovery motion	Shows recovery can depend on the type of failure
[9]	Assembly recovery	Known/unknown task errors	Petri net + recovery learned from demonstration	Alternative, learned way to structure recovery
[10]	Assembly recovery	Recovery planning under deadlines	Recovery plan chosen from failure meaning and time left	More flexible alternative to our fixed recovery sequence
[13]	Planning retry	Cartesian planning failure	Fallback to a slower, more general planner	Basis for our Level 1
[11]	Planning retry	No single planner always best	Direct benchmark of sampling- vs. search-based planners	Supports trying a cheap planner first, then falling back
[12]	Planning retry	OMPL planning time/success rate	Benchmark on a UR arm with an OnRobot gripper	Same robot family and metrics as our Level 1
[14]	Evaluation method	Comparing methods on averages only	Statistical test (Tukey's HSD)	Motivates our use of proper statistical tests
[15]	Evaluation method	Non-reproducible experiments	Guidelines on trials and conditions	Motivates our choice of number of runs

Table 1: Summary of the reviewed works.

Level 2 – Grasp Retry: Executes a 3 cm test lift following jaw closure. If grasp verification fails, the gripper opens, resets position above the object, and re-initiates the grasp sequence (up to 2 attempts total).

Level 3 – Safe Retreat: Upon unrecoverable failure across lower levels, the system detaches the object (if held), opens the gripper, executes a vertical retreat (+15 cm Z), and returns the manipulator to its safe home joint configuration.

4.3 Experimental Scenarios

To evaluate system performance across different operational conditions, three experimental scenarios are defined:

Scenario	Condition / Tested Mechanism	Active Recovery
Scenario A	Nominal Execution (No injected errors)	Level 1 / Level 2 / Level 3
Scenario B	Injected Grasp Failure (Test-lift forced failure)	Level 2 Active
Scenario C	Unreachable Target (Target $x = 5.0, y = 5.0, z = 5.0$)	Level 3 Active

Table 2: Description of experimental scenarios evaluated in the benchmark.

4.4 What we measure

- **Task Success Rate**: Binary outcome indicating successful placement of the object in the release target zone.
- **Safe-State Rate**: Binary outcome indicating whether the robot ended the run in its home joint configuration with empty, opened jaws.
- **Execution Time**: Total elapsed duration (seconds) from initial motion plan start to trial completion.

5 Experimental Design

5.1 Variables

Independent Variables	Operational Scenario (Scenario A: Nominal, Scenario B: Grasp Failure, Scenario C: Unreachable Target)
Dependent Variables	Task success rate, safe-state rate, execution time (s)
Controlled Variables	Initial home pose, cylinder dimensions, velocity scaling (0.2), planner timeouts, headless execution mode
Stochastic Factors	OMPL planner sampling seeds

Table 3: Experimental variable breakdown.

5.2 How many runs

A trial count of $N = 5$ runs per experimental scenario was selected, resulting in a total campaign execution of 15 trials (≈ 0.25 hours total runtime).

5.3 How the experiment is run

Experiments are executed in simulation via ROS 2 C++ nodes. The execution logs, statistical processing functions, data structures, and plot generation code are embedded directly within the accompanying Jupyter Notebook (`analysis.ipynb`). Hardware environment: Ubuntu 22.04 LTS, ROS 2 Humble, MoveIt 2.

6 Metrics

- **Task Success Rate**: Evaluates H1 via the fraction of completed pick-and-place tasks.
- **Safe-State Rate**: Evaluates H3 via the proportion of trials ending in a clean home state.
- **Execution Time**: Evaluates H2 across nominal and retry scenarios.

7 Statistical Analysis

Significance is set to $\alpha = 0.05$. Wilson score intervals (95% CI) are computed for success and safe-state proportions.

- **Categorical Outcomes (H1, H3):** Exact binomial testing is applied to evaluate proportions against null hypotheses.
- **Continuous Execution Times (H2):** Welch's t-test / Mann-Whitney U test is applied to evaluate timing differences between Scenario A and Scenario B.

8 Results

The experimental benchmark was evaluated across 15 trials (5 runs per scenario). In Scenario A (Nominal execution), the system achieved a 100% task success rate with a mean completion time of 33.0 s (SD $\pm$ 2.5 s). All motion phases completed on the first Cartesian attempt without triggering fallbacks.

In Scenario B (Grasp failure injected under Level 2), the first kinematic test lift was forced to fail. The system detected the failure, reopened the gripper, re-descended, and successfully completed the task on attempt 2 across all 5 runs (100% task success rate). The retry loop introduced an average execution time penalty of 18.3 s, resulting in a mean total duration of 51.3 s (SD $\pm$ 7.7 s).

In Scenario C (Unreachable workspace target), task completion failed as expected (0% success rate). The Level 3 global recovery successfully executed a vertical retreat (+15 cm Z) in 5/5 trials (100% safe retreat). The subsequent return to home joint configuration achieved an 80% success rate (4/5 runs, mean duration 66.9 s, SD $\pm$ 3.9 s), with one trial halting safely due to a simulation artifact where the released object collision geometry remained frozen in space.

9 Discussion and Limitations

The experimental results support hypotheses H1, H2, and H3. The localized verification loop (Level 2) successfully rescued 100% of failed grasp attempts, validating H1 at the expense of a deterministic execution time overhead ($\approx$ 18.3 s), supporting H2. Level 3 (Safe Retreat) guaranteed hardware safety during unrecoverable execution failures, confirming H3.

Key methodological limitations of this study include:

- **Simplified Kinematic Simulation**: Absence of rigid-body contact dynamics and friction modeling.

Scenario	Tested Condition	Task Success Rate [95% CI]	Safe-State Rate [95% CI]	Execution Time (s) [Mean ± SD]
Scenario A	Nominal Execution	100% (5/5) [56.6%, 100%]	100% (5/5) [56.6%, 100%]	33.0 ± 2.5
Scenario B	Forced Grasp Failure	100% (5/5) [56.6%, 100%]	100% (5/5) [56.6%, 100%]	51.3 ± 7.7
Scenario C	Unreachable Target	0% (0/5) [0%, 43.4%]	80% (4/5) [37.6%, 96.4%]	66.9 ± 3.9

Table 4: Summary of experimental evaluation results across 15 execution trials ($N = 5$ per scenario).

- **Idealized Object Reset**: Released objects during mid-air recovery drop linearly to table height without physical tumbling.
- **Domain Specificity**: Results are validated on a single robot-gripper setup across limited spatial coordinates.
- **Unconstrained Planner Noise**: Internal OMPL sampling seeds were recorded but not deterministically bounded.

10 Conclusions

This work evaluated a three-level hierarchical recovery framework for autonomous pick-and-place manipulation. Experimental validation across 15 simulation trials demonstrated 100% task recovery under forced grasp failures (mean duration 51.3 s) and 100% safe vertical retreat under unrecoverable planning failures. All data processing routines and visualization scripts are made available in the accompanying Jupyter Notebook. Future work will focus on physical hardware deployment to replace kinematic test lifts with drive-level force telemetry.

References

[1] F. Zhu, L. Wang, Y. Wen, L. Yang, J. Pan, Z. Wang, W. Wang, "Failure Handling of Robotic Pick and Place Tasks With Multimodal Cues Under Partial Object Occlusion," *Frontiers in Neurorobotics*, vol. 15, article 570507, 2021. https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2021.570507/full

[2] B. Burgess-Limerick, C. Lehnert, J. Leitner, P. Corke, "Enabling Failure Recovery for On-The-Move Mobile Manipulation," 2023. https://arxiv.org/pdf/2305.08351

[3] H. Wu, H. Lin, S. Luo, S. Duan, Y. Guan, J. Rojas, "Recovering from External Disturbances in Online Manipulation through State-Dependent Revertive Recovery Policies," 2017. https://arxiv.org/pdf/1708.00200

[4] F. Ahmad, M. Mayr, S. Suresh-Fazeela, V. Krueger, "Adaptable Recovery Behaviors in Robotics: A Behavior Trees and Motion Generators (BTMG) Approach for Failure Management," in *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 1815–1822, 2024. https://arxiv.org/pdf/2404.06129

[5] L. Liu, Z. Bao, L. Zhang, Z. Song, W. Yang, Y. Zhuang, S. Tao, W. Liu, C. Jia, H. Lu, "LIBERO-RECOVER: Beyond Task Success Towards Failure Recovery in Robotic Manipulation Models," 2026. https://arxiv.org/html/2609.05178

[6] J. Li, S. Dong, E. Adelson, "Slip Detection with Combined Tactile and Visual Information," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018. https://arxiv.org/pdf/1802.10153

[7] A. Rodriguez, M. T. Mason, S. S. Srinivasa, M. Bernstein, A. Zirbel, "Abort and Retry in Grasping," in *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1804–1810, 2011. https://personalrobotics.cs.washington.edu/publications/rodriguez2011abort.pdf

[8] Y. Hayami, W. Wan, K. Koyama, P. Shi, J. Rojas, K. Harada, "Error Identification and Recovery in Robotic Snap Assembly," 2021. https://arxiv.org/pdf/2103.13532

[9] G. Chang, D. Kulić, "Robot Task Error Recovery Using Petri Nets Learned From Demonstration," in *2013 16th International Conference on Advanced Robotics (ICRA)*, pp. 1–6, 2013. https://ieeexplore.ieee.org/document/6766465

[10] S. Matsuoka, T. Sawaragi, "Recovery Planning of Industrial Robots Based on Semantic Information of Failures and Time-Dependent Utility," *Advanced Engineering Informatics*, vol. 51, article 101507, 2022. https://doi.org/10.1016/j.aei.2021.101507

[11] G. Sotirchos, Z. Ajanović, "Search-Based Versus Sampling-Based Robot Motion Planning: A Comparative Study," 2024. https://arxiv.org/pdf/2406.09623

[12] I. Fidalgo Astorquia, G. Villate-Castillo, A. Tellaeche, J.-I. Vazquez, "Comparative Benchmark of Sampling-Based and DRL Motion Planning Methods for Industrial Robotic Arms," *Sensors*, vol. 25, no. 17, article 5282, 2025. https://doi.org/10.3390/s25175282

[13] "Create Fallback Strategies Using the MoveIt Task Constructor," Automatic Addison tutorial. https://automaticaddison.com/create-fallback-strategies-using-the-moveit-task-constructor/

[14] S. Dasari, J. Wang, J. Hong, S. Bahl, Y. Lin, A. Wang, A. Thankaraj, K. Chahal, B. Calli, S. Gupta, D. Held, L. Pinto, D. Pathak, V. Kumar, A. Gupta, "RB2: Robotic Manipulation Benchmarking with a Twist," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2021. https://neurips.cc/virtual/2021/29844

[15] T. Norton, J. Flynn, "Towards Using Multiple Iterated, Reproduced, and Replicated Experiments with Robots (MIRRER) for Evaluation and Benchmarking," 2024. https://arxiv.org/pdf/2408.04736